

AlloQuant master results - data dictionary

File: master_kinase_analysis_results_v7r3.csv | 1600 rows x 48 columns | generated 2026-07-30

- This dictionary defines the AlloQuant v7r3 master-CSV SCHEMA. The 48 columns are identical across all v7r3 masters (SRC 260606, SRC 260718 rerun, CDK1-CCNB1); only which rows are populated varies by run.
- Anchored to the SRC 260718 rerun (the AF3-complete master). Empirical counts/ranges below are from that file: 8 systems (CSK apo/holo x SRC apo/holo x SRC WT/pY159) x 25 AF3 models x 2 chains = 1600 rows.
- AF3 confidence columns (ipTM/pTM/PAE_*) are populated in this rerun (it was re-predicted to capture summary_confidences.json/confidences.json). They are N/A in the original SRC 260606 master.
- The CDK1-CCNB1 master is single-chain (kinase only, 600 rows, same 48 columns). Its AF3 columns are not uniformly N/A: pTM 600/600; ipTM 500/600 (N/A only for the ligand-free apo monomer); interface-PAE 400/600 (the CCNB1-containing rows only).
- holo = ATP/(Mg)-bound; apo = no nucleotide. C_Spine == 'No Ligand' flags an apo chain.
- Distances in Angstrom (A), angles in degrees. Landmark indices are 0-based HMM-aligned positions; numbered landmarks (n99, v104, k105, e107, m118, m120, e121, i150, y156, d220) are labelled in PKA numbering (the canonical reference kinase), and map to the equivalent target position via the HMM regardless of the residue type actually present.
- Missing / uninformative values: geometry that fails to compute is written as the sentinel 999.9; everything else uninformative is the string 'N/A' (or 'None' for CoFactor_Name / Partner). Downstream R reads "", 'NA', 'N/A', 'None' all as NA.
- Writers: columns 1-43 by modules/chimerax_hmm_worker_v7r3.py; columns 44-48 (ipTM, pTM, PAE_*) by modules/extract_af3_metrics.py, joined on Directory.
- v7r2 -> v7r3 (2026-07-29) CHANGES REPORTED VALUES: re-run any dataset before comparing across versions. Three Module-1 corrections, below. Dihedral, D2_Dist, SB_Dist and every other column are unchanged.
- (i) D1_Dist is now measured from the alphaC-Glu(+4) Calpha per Modi & Dunbrack, PNAS 2019 (116:6818), not the alphaC-Glu itself; the published 11/14 A cutoffs are calibrated on the (+4) anchor. This also changes Spatial and State.
- (ii) the ActLoop_CT anchor is the HRD arginine (hrd+1), not hrd+6, which is not a conserved position (an Arg in CSK but an Ala in SRC).
- (iii) a missing-density guard makes ActLoop_NT/ActLoop_CT return N/A across unresolved residues instead of measuring over the gap; no effect on complete AlphaFold3 chains.

#	Column	Group	Type	Units	Populated	Allowed values / range	Empirical (this CSV)	Description
1	Simulation_ID	Identity	string		always		filled 1600/1600	Unique model id, from the model path via get_core_name() (dedups repeated path tokens). Encodes the two chains, AF3 seed and sample, e.g. a-csk-wtcat-apo_b-src-wtcat-holo_seed-..., sample-N_model.
2	Directory	Identity	string (path)		always		filled 1600/1600	Relative path to the AF3 prediction folder (system / seed-sample). Also the join key used to merge in the AF3 confidence columns.
3	File	Identity	string		always	model.cif	filled 1600/1600	Structure file analysed within Directory (AF3 predicted mmCIF).
4	Chain	Identity	categorical		always	A, B, D	filled 1600/1600; observed: A 800, B 400, D 400	mmCIF chain id of the protein chain in this row. A = CSK protomer; B or D = SRC protomer (the letter shifts to D when holo ligand chains are inserted ahead of it). One row per protein chain; 2 rows per model.
5	Type	Identity	categorical		always	CSK-WT, SRC-WT	filled 1600/1600; observed: CSK-WT 800, SRC-WT 800	Kinase identity assigned by best sequence-identity match to the FASTA landmark set (get_best_landmark_for_chain). Note the SRC pY159 variant still matches as SRC-WT; the pY159 vs WT distinction lives in Simulation_ID/Directory, not here.
6	State	Conformation	categorical		always	see Description	filled 1600/1600; observed: Active (BLAminus) 1063, Inactive (BLBplus) 524, Inactive (BLBminus) 5, Inactive (BLAplus) 5, Inactive (BLBtrans) 1, Active-Like (ABAminus) 1, DFGinter (BABplus) 1	Human-readable state = Spatial + Dihedral with an activity gloss: Active (BLAminus), Active-Like (ABAminus), Inactive (BLBplus / BLBtrans / BLAplus / BLBminus / BBABminus / BABtrans), else a fall-through <spatial> (<dihedral>) such as DFGinter (BABplus). Downstream analysis treats 'Active (BLAminus)' as the active target.
7	Role	Interface	categorical		always (dimer)	C-lobe Donor, N-lobe Receiver, Symmetric, Unpaired, Co-factor Bound	filled 1600/1600; observed: N-lobe Receiver 799, C-lobe Donor 799, Unpaired 2	Role in the asymmetric dimer from analyze_dimer_interface: a chain is C-lobe Donor / N-lobe Receiver when its C-lobe contacts the partner N-lobe within 8 A (Symmetric if mutual, Unpaired if no contact). For CSK-SRC: CSK=N-lobe Receiver, SRC=C-lobe Donor.
8	Partner	Interface	categorical		always (dimer)	A, B, D	filled 1598/1600, blank 2; observed: A 799, B 400, D 399	Chain id of this chain's interface partner (or None if Unpaired; a cofactor name for Co-factor Bound systems).
9	CoFactor_Name	Cofactor	string		N/A (no cofactor)		filled 0/1600, blank 1600	Name of the nearest non-kinase chain within 15 A, else None. Populated only for cofactor systems (CDK-cyclin, RAF/14-3-3, etc.); empty for the CSK-SRC pair.
10	CoFactor_aC_Dist	Cofactor	float	A	N/A (no cofactor)		filled 0/1600, blank 1600	Min distance from the kinase alphaC region (c-4..c+4) to the cofactor chain. Cofactor systems only.
11	CoFactor_ActLoop_Dist	Cofactor	float	A	N/A (no cofactor)		filled 0/1600, blank 1600	Min distance from the kinase activation loop (f..APE) to the cofactor chain. Cofactor systems only.
12	Interface_C_Lobe_Donor_Dist	Interface	float	A	always (dimer)	2.44 to 13.24	filled 1598/1600, blank 2	Min distance between chain-A C-lobe and chain-B N-lobe atoms (donor face). 999.9 sentinel if lobes cannot be split; N/A if unpaired.
13	Interface_N_Lobe_Rec_Dist	Interface	float	A	always (dimer)	2.44 to 13.24	filled 1598/1600, blank 2	Min distance between chain-B C-lobe and chain-A N-lobe atoms (receiver face). Same sentinel/NA conventions.
14	R_Spine	Spine/helix	categorical		always	Intact, Broken, Missing	filled 1600/1600; observed: Intact 1447, Broken 153	Regulatory-spine integrity: Intact if all three R-spine contacts (HRD-His-DFG-Phe, DFG-Phe-rs1, rs1-rs2) are <4.5 A, else Broken (Missing if landmarks absent).
15	C_Spine	Spine/helix	categorical		always	Intact, Ligand Distant, No Ligand	filled 1600/1600; observed: No Ligand 800, Intact 800	Catalytic-spine vs adenine: Intact if the VAIK region (k..k-3) is <6 A from a bound ligand, Ligand Distant if >=6, No Ligand if apo. Serves as the apo/holo discriminator downstream (No Ligand = apo chain).
16	C_Helix	Spine/helix	categorical		always	In, Out	filled 1600/1600; observed: In 878, Out 722	alphaC rotamer: In if the VAIK-Lys CB to alphaC-Glu CB distance is <=10 A (salt-bridge competent), else Out. In = active-like.
17	Shell_State	Spine/helix	categorical		always	Packed, Loose	filled 1600/1600; observed: Packed 1600	Regulatory-shell packing: Packed if Shell_M118_M120_Dist <5 A, else Loose (only Packed observed in this run).
18	Spatial	Conformation	categorical		always	DFGin, DFGout, DFGinter, Outlier	filled 1600/1600; observed: DFGin 1599, DFGinter 1	Dunbrack spatial DFG group from D1_Dist/D2_Dist: DFGin (D1<=11 & D2>=11), DFGout (D1>11 & D2<=14), DFGinter (both <=11), else Outlier. All four are reachable; correcting the D1 anchor in v7r3 removed the inflated Outlier/DFGout calls seen in v7r2.
19	Dihedral	Conformation	categorical		always	BLAminus, BLBplus, ...	filled 1600/1600; observed: BLAminus 1063, BLBplus 524, BLBminus 5, BLAplus 5, BLBtrans 1, ABAminus 1, BABplus 1	KinCore backbone cluster = Ramachandran regions of X(f-2), Asp(f-1), Phe(f) + DFG-Phe chi1 rotamer, e.g. BLAminus/ABAminus/BLBplus/BLBtrans/BLAplus/BLBminus. Region codes A/B/L/E; rotamer plus/trans/minus. BLAminus = canonical active.

- v7r2 -> v7r3 (2026-07-29) CHANGES REPORTED VALUES: re-run any dataset before comparing across versions. Three Module-1 corrections, below. Dihedral, D2_Dist, SB_Dist and every other column are unchanged.

20	ActLoop_NT	Conformation	categorical		always	NTin, NTout, N/A	filled 1600/1600; observed: NTin 1170, NTout 430	Activation-loop N-terminal packing: NTin if residues f+3..f+6 come within 5.5 Å of the residue before HRD (hrd-1), else NTout; N/A if those residues are not contiguous in the deposited numbering (missing-density guard).
21	ActLoop_CT	Conformation	categorical		always	CTin, CTout, N/A	filled 1600/1600; observed: CTin 1564, CTout 36	Activation-loop C-terminal packing: CTin if the pre-APE window (ape-6 down to ape-9, floored at f+2) comes within 5.5 Å (all-atom) of the HRD arginine (hrd+1), else CTout; N/A if the residues are not contiguous in the deposited numbering. The hrd+1 anchor follows Modi & Dunbrack's APE9-Arg contact (v7r2 used hrd+6, which is an Arg in CSK but an Ala in SRC); only the anchor changed at v7r3, the window and the 5.5 Å all-atom cutoff being inherited unchanged from v6r6. NOT COMPARABLE WITH KINCORE'S ActLoopCT LABEL: Kincore measures two named atoms of the single APE9 residue (CA to the Arg backbone O) at 6.0 Å, whereas this is an all-atom minimum over four residues at 5.5 Å. The 5.5 Å cutoff is stricter than Kincore's 6.0 Å; it is the four-residue window, not the cutoff, that makes this the more permissive criterion. Because the value is an all-atom minimum over whatever side chains occupy the window, it is also sequence-dependent across kinases.
22	Phi_D	Conformation	float	degrees	always	-100.82 to 70.18	filled 1600/1600	Backbone phi dihedral of the DFG-Asp (f-1).
23	Psi_D	Conformation	float	degrees	always	15.46 to 113.62	filled 1600/1600	Backbone psi dihedral of the DFG-Asp (f-1).
24	Cleft_Gape_Dist	Active site	float	Å	always	11.65 to 19.61	filled 1600/1600	Inter-lobe cleft opening: roof Calpha to floor Calpha, where roof = a P-loop residue (phospho/Tyr/Thr/Ser or P-loop midpoint) and floor = catalytic HRD-Asp (hrd+2). Geometric, so populated for apo and holo.
25	Mg_Hijack_Dist	Active site	float	Å	holo + Mg only	7.64 to 64.16	filled 1200/1600, blank 400	Min distance from the roof residue phosphate/hydroxyl atoms to any Mg2+ ion. Requires a bound Mg2+; N/A on apo chains.
26	Substrate_Clearance_Angle	Active site	float	degrees	nucleotide-bound only	88.33 to 158.70	filled 800/1600, blank 800	Angle roof-Calpha / ATP-phosphate / floor-Calpha (phosphate atom as vertex). Requires a bound nucleotide; N/A on apo chains.
27	D1_Dist	Active site	float	Å	always	4.69 to 9.89	filled 1600/1600	Dunbrack D1 coordinate: alphaC-Glu(+4) Calpha to DFG-Phe CZ (farthest side-chain atom if CZ absent). The (+4) anchor is the one the 11/14 Å Spatial cutoffs are calibrated on (Modi & Dunbrack, PNAS 2019, 116:6818); v7r2 measured from the alphaC-Glu itself, which compressed the range.
28	D2_Dist	Active site	float	Å	always	7.80 to 16.74	filled 1600/1600	Dunbrack D2 coordinate: VAIK-Lys Calpha to DFG-Phe CZ.
29	SB_Dist	Active site	float	Å	always	2.44 to 18.43	filled 1600/1600	beta3 Lys - alphaC Glu salt bridge: min distance from VAIK-Lys NZ to the nearest polar (O*/N*) side-chain atom of the alphaC-Glu. Short (~2.5-3 Å) = intact/active.
30	HRD_ATP_Dist	Active site	float	Å	holo only	2.47 to 8.92	filled 800/1600, blank 800	Min distance from catalytic HRD-Asp (hrd+2) side-chain O*/N* to the ATP gamma/beta/alpha-phosphate group. Requires a bound nucleotide; N/A on apo chains.
31	DFG_Mg_Dist	Active site	float	Å	holo + Mg only	0.91 to 6.97	filled 796/1600, blank 804	Min distance DFG-Asp (f-1) side-chain O*/N* to the nearest Mg2+ (recorded only if <=15 Å). Requires Mg2+; N/A on apo chains.
32	DFG_ATP_Dist	Active site	float	Å	holo only	1.26 to 4.70	filled 800/1600, blank 800	Min distance DFG-Asp side-chain O*/N* to ATP phosphate atoms. Requires a bound nucleotide; N/A on apo chains.
33	PLoop_ATP_Dist	Active site	float	Å	holo only	1.44 to 5.22	filled 800/1600, blank 800	Min distance from P-loop backbone/CB atoms to ligand phosphate atoms. Requires a bound nucleotide; N/A on apo chains.
34	aCb4_aE_Dist	Allosteric	float	Å	always	2.81 to 3.72	filled 1600/1600	Min polar-atom (N*/O*) distance between the alphaC-beta4 loop (c+8..c+14) and the alphaE helix (hrd-25..hrd-10). Allosteric relay bridge.
35	Spine_Bridge_Dist	Allosteric	float	Å	always	3.57 to 8.36	filled 1600/1600	alphaC-beta4 loop to ligand if a ligand is bound; otherwise (apo) loop to k-3 residue. Populated for both apo and holo (fallback for apo).
36	V104_RS2_Dist	Allosteric	float	Å	always	6.81 to 7.38	filled 1600/1600	Min side-chain distance V104 to R-spine residue rs2 (beta4).
37	I150_HRD_Dist	Allosteric	float	Å	always	9.65 to 10.51	filled 1600/1600	Min side-chain distance I150 to the catalytic HRD-His.
38	Shell_M118_M120_Dist	Allosteric	float	Å	always	3.57 to 4.45	filled 1600/1600	Min side-chain distance between the two hydrophobic-shell positions M118 and M120 (PKA numbering; Ile79/Thr81 in CSK and SRC, Leu78/Phe80 in CDK1); drives Shell_State.
39	Y156_N99_Dist	Allosteric	float	Å	always	7.87 to 9.86	filled 1600/1600	Residue-min distance Y156 to N99 (alphaE anchor).
40	K105_E107_Dist	Allosteric	float	Å	always	2.95 to 5.71	filled 1600/1600	Min side-chain distance K105 to E107.
41	K105_E121_Dist	Allosteric	float	Å	always	2.97 to 4.16	filled 1600/1600	Min side-chain distance between residues at Pfam Pkinase HMM node 62 (PKA canonical K105 position; GLN65 in CSK domain numbering) and HMM node 78 (PKA canonical E121 position; GLU82 in CSK domain numbering). Part of the alphaC-beta4 loop contact network. NOT the beta3-Lys/alphaC-Glu regulatory salt bridge -- for that, see SB_Dist.
42	K105_N99_Dist	Allosteric	float	Å	always	2.72 to 7.11	filled 1600/1600	Min side-chain distance K105 to N99.
43	D220_HRD_Dist	Allosteric	float	Å	always	3.35 to 3.84	filled 1600/1600	Min side-chain distance D220 to the catalytic HRD-His.
44	ipTM	AF3 confidence	float	0-1	dimers	0.32 to 0.90	filled 1600/1600	AF3 interface pTM from summary_confidences.json (>0.8 high, 0.6-0.8 moderate, <0.6 low). N/A for single-entity/monomer models. Populated in this rerun; N/A in the original 260606 master.
45	pTM	AF3 confidence	float	0-1	always	0.57 to 0.91	filled 1600/1600	AF3 predicted TM-score. Populated in this rerun; N/A in the original 260606 master.
46	PAE_A_to_B	AF3 confidence	float	Å	dimers	8.40 to 27.45	filled 1600/1600	Mean predicted aligned error over the [chain-A rows, chain-B cols] PAE block (<5 rigid, >15 non-interacting). N/A for monomers.
47	PAE_B_to_A	AF3 confidence	float	Å	dimers	7.99 to 26.56	filled 1600/1600	Mean PAE over the transposed [chain-B rows, chain-A cols] block. N/A for monomers.
48	PAE_Mean_AB	AF3 confidence	float	Å	dimers	8.43 to 27.00	filled 1600/1600	Symmetric interface PAE = mean of PAE_A_to_B and PAE_B_to_A. N/A for monomers.